Towards Computational Phase Imaging for Monitoring Light-Based Fabrication

Zhiping Wang[1], Felix Wechsler[1], Jonathan Dong[2], Christophe Moser[1]

[1]Laboratory of Applied Photonics Devices, EPFL, Switzerland

[2]Biomedical Imaging Group, EPFL, Switzerland

Poster contact: zhiping.wang@epfl.ch

Personal website

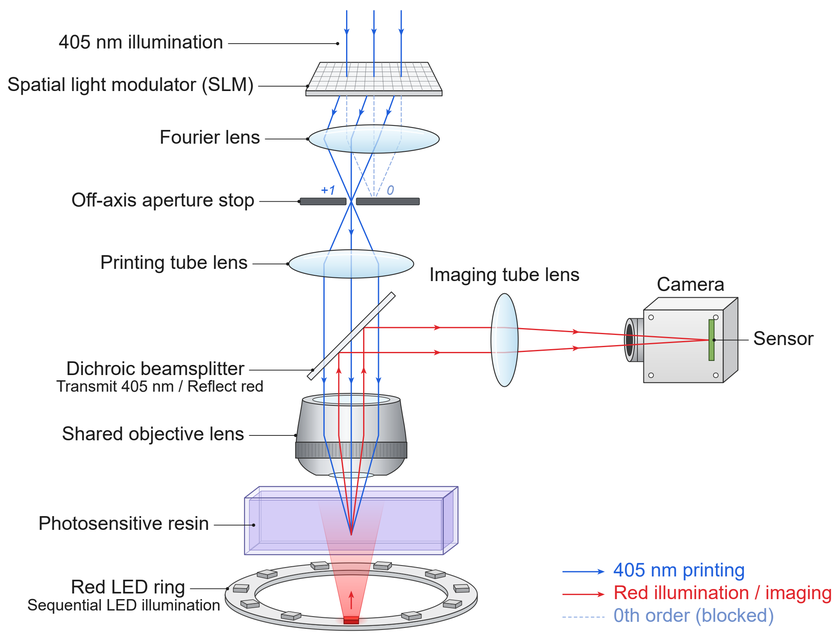
Light-based fabrication is an important route to 3D manufacturing.

Yet observing the structure **as it forms remains difficult**.

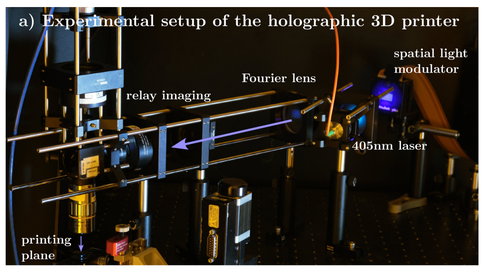
We explore **computational phase imaging** for in situ monitoring.

Integrating phase imaging into SHVAM

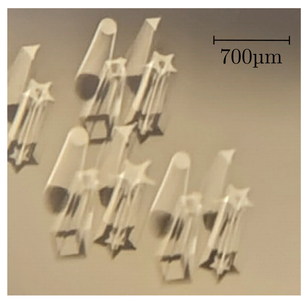
Single-View Holographic Volumetric Additive Manufacturing (SHVAM) [1] prints a whole part at once using 405 nm holograms. **We propose an FPM imaging path:** a red LED ring illuminates the print; a dichroic directs the red light through the shared objective to a camera while transmitting the printing light.



Proposed printing + imaging path: blue printing and red imaging share the objective.



SHVAM apparatus [1]



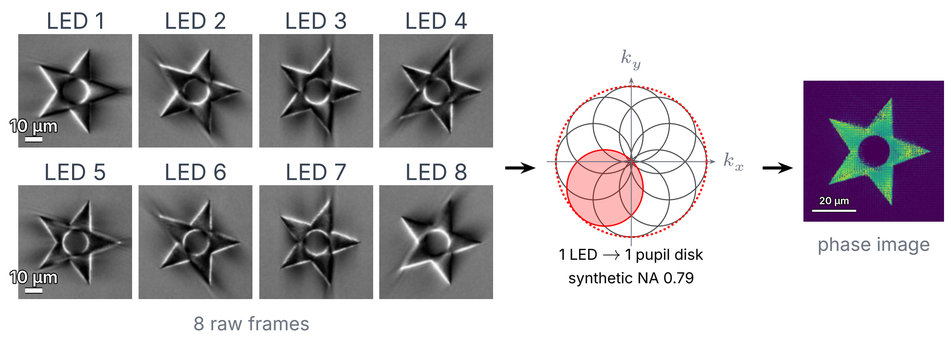
SHVAM prints [1]

Why phase? Fourier ptychography

Liquid and cured resin differ by a small Δn, producing a phase delay $\varphi = (2\pi/\lambda)\int \Delta n \, \mathrm{d}z$. **Fourier ptychographic microscopy (FPM)** [2] retrieves phase from sequential LED images:

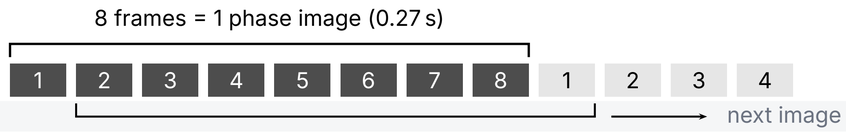
$$I_j = \left|\mathcal{F}^{-1}\{P(\mathbf{k})\tilde{O}(\mathbf{k}-\mathbf{k}_j)\}\right|^2$$

$$\min_O \sum_j \left\| \sqrt{I_j^{\text{meas}}} - \sqrt{I_j(O)} \right\|^2 + \tau\,\mathrm{TV}(O)$$



8 raw frames

Sequential acquisition. Eight annular LEDs [3] at 30 fps; each estimate uses an overlapping 8-frame window (0.27 s acquisition). Compute time is separate.



The simulated SHVAM print

A star with a central hole, printed by **20 SHVAM phase holograms, cycled on the SLM for 8 s**; SHVAM model, $\Delta n = 0.01$.

Until 7.2 s: the light uses up the dissolved inhibitors; no phase yet.

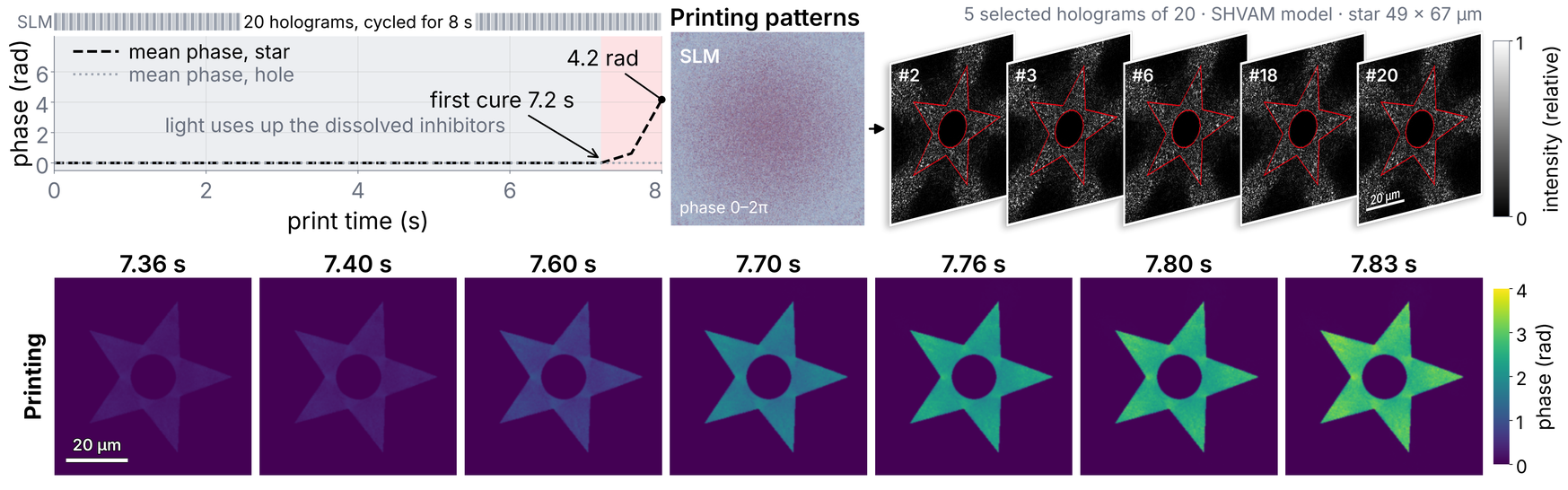
Then the whole star cures within 0.8 s.

Each hologram puts 15 % of its power on the star; the rest stays below threshold.

7.2 s first cure

0.8 s cure

4.2 rad final phase



FPM tracking in simulation

8 LEDs at 30 fps, simulated camera with noise. Reconstructions use overlapping 8-frame windows (0.27 s acquisition), shifted by one frame.

Reconstruction: warm-started L-BFGS (50 iterations per window), region-anchored hybrid unwrapping; processing latency is separate from acquisition timing.

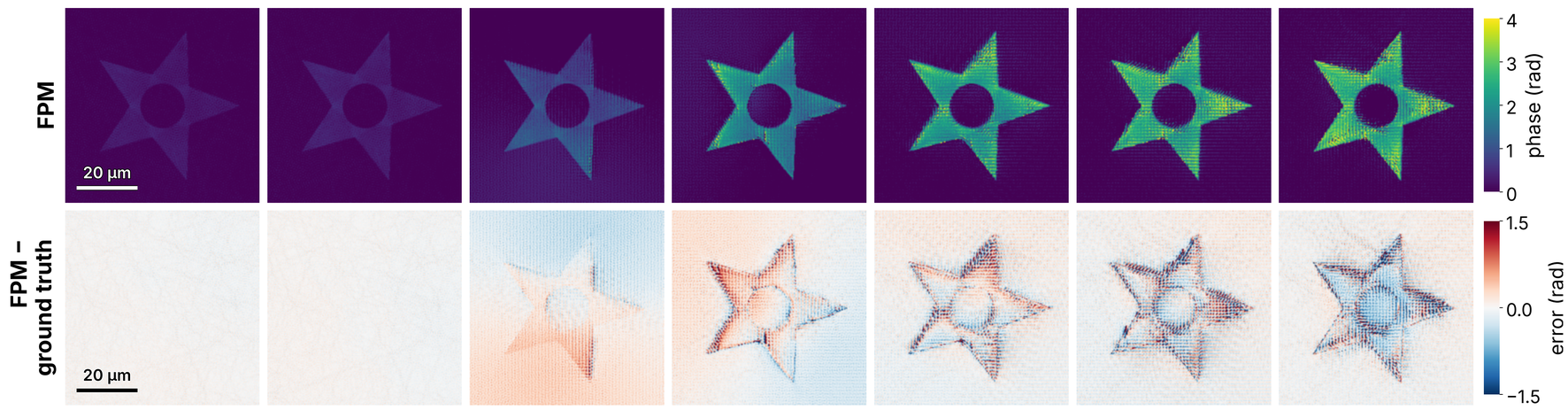
Result: In simulation, FPM follows the cure, with no false phase before curing starts. Largest error of the mean phase in the star: 0.25 rad (lag during the fast cure).

Next: validate on real prints.

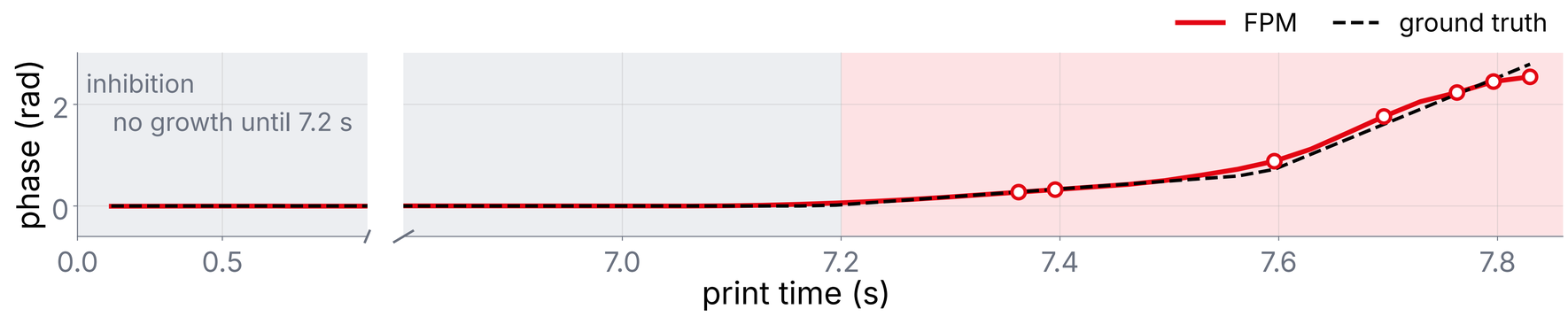
0.25 rad largest error

0.27 s per image

0.01 rad before cure



Printing (above), FPM and FPM minus ground truth at the same seven times; circles in the plot below mark them. Each row shares its scale bar.



References

1. Wechsler, F., Rizzo, R., and Moser, C. (2026). Single-View Holographic Volumetric 3D Printing with Coupled Differentiable Wave-Optical and Photochemical Optimization. *ACM Transactions on Graphics*. arXiv:2601.16330.
2. Zheng, G., Horstmeyer, R., and Yang, C. (2013). Wide-field, high-resolution Fourier ptychographic microscopy. *Nature Photonics*, 7, pp. 739–745.
3. Sun, J., Zuo, C., Zhang, J., Fan, Y., and Chen, Q. (2018). High-speed Fourier ptychographic microscopy based on programmable annular illuminations. *Scientific Reports*, 8, p. 7669.